ADDITIONAL GUIDE

How to enable the onboard TFT display on the STM32F429I-DISC1 board in the LAB6 project

This guide is intended to help students quickly reuse a working display setup in a new laboratory project. The practical goal is simple: keep the LAB6 I2C logic focused on the assignment itself, while showing the communication results on the onboard TFT display in the same way as in the LAB5 project.

Main point: on the STM32F429I-DISC1 board the display does not work from `LTDC` alone. It requires the full set: `LTDC + FMC/SDRAM + DMA2D + SPI5 + LCD control GPIO`. The fastest and safest approach is not to rebuild everything from scratch, but to reuse the already working display base from LAB5.

1. What must remain enabled in the CubeMX project

If the display has to work, the following blocks must remain enabled in the CubeMX project:

- `GPIO`
- `DMA2D`
- `FMC` with SDRAM configuration
- `LTDC`
- `SPI5`
- `RCC` and `SYS`

- PG13 and PG14 if LED status indication is still needed

Important pins for LCD control:

- PF7 / PF8 / PF9 – SPI5_SCK / SPI5_MISO / SPI5_MOSI
- PC2 – LCD CS
- PD12 – LCD RDX
- PD13 – LCD WRX
- all LTDC and FMC pins generated by CubeMX for the display and SDRAM

Note: I2C3 is not required for the TFT display itself. In LAB6, I2C3 is required for the laboratory task, not for LCD initialization.

2. Which files are most practical to copy from a working LAB5 project

If a new project is being created, it is recommended to copy the following files from the LAB5 project:

2.1. From Core/Src

- ili9341.c
- stm32f429i_discovery.c
- stm32f429i_discovery_lcd.c
- stm32f429i_discovery_sdram.c
- font8.c
- font12.c
- font16.c
- font20.c
- font24.c

2.2. From `Core/Inc`

- `ili9341.h`
- `lcd.h`
- `stm32f429i_discovery.h`
- `stm32f429i_discovery_lcd.h`
- `stm32f429i_discovery_sdram.h`
- `fonts.h`

If these files are not added, the two most common outcomes are: the project fails to link because of missing symbols, or the display remains black because low-level initialization is missing.

3. Required initialization order in `main.c`

The display startup order should match the working LAB5 project: first initialize the required peripherals, then call the LCD initialization functions.

```
#include "stm32f429i_discovery_lcd.h"


MX_GPIO_Init();
MX_DMA2D_Init();
MX_FMC_Init();
MX_LTDC_Init();
MX_SPI5_Init();


if (BSP_LCD_Init() != LCD_OK)
{
    Error_Handler();
```

```
}


BSP_LCD_LayerDefaultInit(0U, LCD_FRAME_BUFFER);
BSP_LCD_SelectLayer(0U);
BSP_LCD_DisplayOn();
BSP_LCD_SetFont(&Font12);
BSP_LCD_SetBackColor(LCD_COLOR_BLACK);
BSP_LCD_Clear(LCD_COLOR_BLACK);
BSP_LCD_SetTextColor(LCD_COLOR_WHITE);
BSP_LCD_DisplayStringAt(0U, LINE(2), (uint8_t *)"LAB6 display works", CENTER_MODE);
```

After this sequence, text should appear on the display. If it does, the display framework is already working, and the LAB6 I2C logic can be integrated safely.

4. What is worth reusing from LAB5 as a reference

- **`main.c`** – shows the correct order of `MX_GPIO_Init()`, `MX_DMA2D_Init()`, `MX_FMC_Init()`, `MX_LTDC_Init()` and `MX_SPI5_Init()`.
- **`lab5_demo.c`** – shows a short and practical LCD startup sequence using `BSP_LCD_Init()`, `BSP_LCD_LayerDefaultInit()`, `BSP_LCD_SelectLayer()` and `BSP_LCD_Clear()`.
- **`stm32f429i_discovery_lcd.c`** – contains the higher-level LCD driver.

- **stm32f429i_discovery.c** – contains low-level SPI5 and LCD control GPIO initialization.
- **stm32f429i_discovery_sdram.c** – initializes the SDRAM required for the LCD frame buffer.

5. Practical workflow for the LAB6 project

1 Create the LAB6 project with the required I2C3 logic.

2 Copy the display BSP files and font files from a working LAB5 project.

3 Do not remove DMA2D, FMC, LTDC and SPI5 from CubeMX.

4 Keep the same LCD initialization order in main.c.

5 First display one simple test string, for example "LAB6 display works".

6 Only after the test string is shown correctly, integrate the LAB6 byte transmission and reception output.

6. Most common mistakes

- **Black screen:** most often FMC, LTDC or SPI5 was removed, or the initialization order was broken.
- **Project fails to link:** most often ili9341.c, stm32f429i_discovery_lcd.c or font12.c and other font files are missing.

- **The display stops working after CubeMX regenerate:** required pins or peripherals were removed from the `.ioc` file.
- **HardFault after LCD startup:** most often SDRAM was not initialized or the frame buffer address is wrong.

7. Recommendation for this course

If the goal is not to study low-level LCD configuration itself, but to show LAB6 I2C results on the display, then the technically correct solution is to **reuse the already working LAB5 display base as a template** and place the LAB6 logic on top of it. This reduces the risk of breaking the display and saves a significant amount of time.